Personal Statement

Nanyang Technological University, Singapore
Jianming Xing MSc in Robotics and Intelligent Systems

On a Dobot CR5, I learned that faster-looking execution can be less reliable. In my tests, non-blocking execution increased apparent command frequency, but a new policy inference could use an intermediate in-motion arm state before the preceding action chunk had finished. The resulting corrections drove the robot back and forth. I therefore used blocking, short-horizon ServoP action chunks, accepting lower apparent frequency to preserve state-action consistency and trajectory quality. This sharpened my guiding question: how can learning, control, timing, sensing, and safety form one reliable robot system?

My preparation combines complementary disciplines by design. In June 2025, I earned a B.Eng. in Mechanical Design, Manufacturing and Automation at Harbin Institute of Technology, Weihai. That degree and robotics projects grounded me in kinematics, robot modeling, and physical constraints. I am now pursuing a second bachelor's degree in Computer Science and Technology at Harbin Institute of Technology, Shenzhen, expected in June 2027. This is not a departure from mechanical engineering, but a deliberate expansion from modeling machines to building the algorithms, data pipelines, and interfaces through which they perceive, learn, and act. Together, these perspectives let me examine embodied systems from physical and computational sides.

That combination became concrete in a CR5 data-collection and inference loop using an Orbbec Astra2 RGB-D camera, an electric gripper, and ROS2. I made RGB timestamps the HDF5 master clock, aligning robot states, target actions, and gripper feedback through binary search and linear interpolation. Fast DDS shared memory and RGB-only capture reduced recorded frame drops from 27.15% to the 2.57%-4.74% range. Beyond throughput alone, I checked schema completeness and count consistency, position and rotation jumps, gripper changes, workspace bounds, and per-step motion, then reviewed an episode in slow playback before further use or replay. I then converted synchronized episodes to LeRobot v2.0 and adapted OpenPI for CR5, using consistent seven-dimensional state and action semantics across collection, training configuration, and WebSocket-served deployment. The client translated policy chunks into ServoP arm targets and Modbus gripper commands. This continuity mattered because a dataset field is not merely a storage choice: it defines what the policy learns to predict and what the hardware later interprets. The project therefore made reliability a chain of aligned decisions across clocks, representations, validation, and execution timing.

A complementary research thread examines action representation. As a Research Assistant at iLearn, I am second author of Hermite Action Tokenizer for VLA, an unsubmitted working paper, and I am responsible for its discrete autoregressive tokenization branch. Working on this branch made representation a control question: what trajectory structure should tokenization preserve, and how should gripper decisions be encoded separately? The method separates seven-dimensional action chunks into six-dimensional end-effector trajectories and gripper states, with shared breakpoints removing endpoint redundancy and quantization seams. Under this offline protocol, which averages four LIBERO subsets with 8-by-7 action chunks, Hermite's Endpoint MSE was approximately 99.7% lower than FAST and 82.6% lower than BEAST. These results show codec fidelity under the stated conditions; they do not establish online rollout performance or physical-robot error. The research showed that action encoding constrains what a model can express; CR5 deployment showed that execution schedules determine which state the model observes and what its command controls. Reliable embodied intelligence therefore depends on representation and execution being joined through synchronized data.

During the Xbotics Embodied AI Community Internship, I migrated Isaac Lab's ANYmal-C navigation task into a MotrixLab NumPy environment. My judgment was to treat migration as interface preservation, not transcription. I refactored reset and step flows, command sampling, observations, rewards, and termination, corrected MuJoCo heightfield frames, added spawn and goal constraints, and used reward and termination regression checks. Earlier, in the HIT Weihai HERO RoboMaster Team's Vision Group, I helped modularize image acquisition, vision processing, and communication synchronization as ROS2 nodes and worked on host-MCU timing. These experiences reinforced that coordinate, interface, and timing assumptions must be explicit for integration.

My immediate goal is an R&D engineering role in robotics or embodied AI, connecting learning methods to sensing, control, and deployment. I want to strengthen my ability to design and evaluate real-hardware systems, where representation, timing, and physical constraints must be reconciled. After gaining stronger real-system experience, I may pursue doctoral study; it remains an option rather than a predetermined destination. I therefore seek graduate training whose curriculum and applied opportunities can close gaps between my integration experience and the robot systems I want to build.

Migrating an ANYmal-C navigation task made me confront how commands, observations, rewards, and termination together define mobile-robot autonomy. My CR5 work raised a complementary interface problem for manipulation: visual state and policy actions must remain meaningful while the arm moves. My separate Hermite research adds a third layer: how an offline trajectory representation defines what an interface may eventually need to carry. I now want to understand how multimodal perception, task definitions, and control should change between mobile robots and manipulators. NTU's MSc in Robotics and Intelligent Systems addresses that systems-level question.

If its resources remain available for my target intake, the published January 2027 curriculum offers a path into that question. MA6214 Autonomous Mobile Robot would extend my command and observation work into multimodal fusion, planning, and motion control. MA6212 Robotics Manipulation and Advanced Control would let me compare mobile-robot interfaces with manipulation, while MA6211 AI and Machine Learning Fundamentals for Robotics would help me examine how learned action representations meet both sets of task definitions. If approved, MA6215 Robotics Projects in the coursework-only route could support a prototype comparing how one representation interfaces with mobile planning and manipulator execution under different observation and completion rules. I would contribute experience from navigation migration, offline action-tokenizer research, and CR5 deployment, then apply the comparison to autonomous-system R&D.